

GURVANTES, Mongolia

WMO: 443740

Lat: 43.2310N

Lon: 101.0430E

Elev: 1730

StdP: 82.20

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.9	-33.5	0.2	-16.1	-30.9	0.3	-16.9	14.4	-11.1	12.7	-9.2	2.0	230	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	30.4	13.7	28.9	13.0	27.5	12.5	16.1	24.1	15.0	23.6	14.2	23.5	4.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.8	12.1	18.3	12.1	10.9	18.1	10.6	9.8	18.2	51.9	24.0	48.1	23.5	45.6	23.7	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.3	10.4	-26.3	32.6	2.4	1.5	-28.1	33.7	-29.5	34.6	-30.8	35.4	-32.6	36.5		
(5)			-27.0	17.4	2.4	2.0	-28.7	18.8	-30.1	20.0	-31.4	21.1	-33.1	22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-13.0	-9.0	-0.8	7.8	13.6	19.5	22.1	20.4	14.0	6.0	-3.4	-10.9	(6)
(7)		DBStd	13.01	4.91	5.77	5.73	5.27	4.62	3.45	2.97	3.11	3.99	4.84	4.74	4.67	(7)
(8)		HDD10.0	2899	712	533	334	99	20	0	0	0	12	139	402	648	(8)
(9)		HDD18.3	4922	971	766	592	317	155	27	5	16	134	382	652	907	(9)
(10)		CDD10.0	1297	0	0	1	33	133	284	375	324	133	15	0	0	(10)
(11)		CDD18.3	278	0	0	0	1	10	60	122	81	6	0	0	0	(11)
(12)		CDH23.3	2098	0	0	0	2	72	474	990	534	25	0	0	0	(12)
(13)		CDH26.7	497	0	0	0	0	8	107	276	106	1	0	0	0	(13)
(14)	Wind	WSAvg	4.4	3.1	3.9	5.1	5.7	5.8	4.7	4.3	4.0	4.0	4.4	4.4	3.6	(14)
(15)	Precipitation	PrecAvg	97	1	2	5	4	7	15	26	25	13	4	2	2	(15)
(16)		PrecMax	173	5	8	21	17	35	93	96	71	47	13	7	12	(16)
(17)		PrecMin	36	0	0	0	0	0	0	2	0	0	0	0	0	(17)
(18)		PrecStd	41	1	2	6	5	9	18	20	19	13	4	2	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	6.2	14.9	22.8	27.7	31.6	33.0	31.2	25.7	19.9	10.6	1.6	(19)
(20)			MCWB	-3.4	-1.0	4.1	8.4	10.3	13.3	15.1	14.6	10.5	7.4	2.3	-3.7	(20)
(21)		2%	DB	-1.4	3.2	12.2	19.8	25.3	29.3	31.1	29.1	23.8	17.5	8.1	-1.1	(21)
(22)			MCWB	-5.5	-2.9	2.2	6.7	9.6	12.3	14.2	13.5	9.9	6.4	0.8	-5.4	(22)
(23)		5%	DB	-3.8	1.2	9.9	18.0	23.4	27.7	29.5	27.7	22.3	15.2	5.8	-2.8	(23)
(24)			MCWB	-7.2	-4.0	1.0	5.8	8.8	11.7	13.5	12.7	9.3	5.2	-0.6	-6.4	(24)
(25)		10%	DB	-5.8	-0.8	7.7	16.0	21.4	26.1	28.0	26.3	20.7	13.4	3.8	-4.5	(25)
(26)			MCWB	-8.6	-5.2	-0.2	4.7	7.9	11.2	13.1	12.5	8.7	4.1	-1.7	-7.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.1	-0.7	4.4	9.1	11.2	14.9	18.0	18.0	12.5	8.1	2.7	-3.7	(27)
(28)			MCDWB	1.4	5.5	13.5	20.7	25.4	23.4	26.4	24.6	18.4	18.8	9.8	0.9	(28)
(29)		2%	WB	-5.3	-2.5	2.6	7.0	10.0	13.9	16.5	16.2	11.5	6.8	1.0	-5.2	(29)
(30)			MCDWB	-1.6	3.1	11.3	18.7	23.2	23.6	24.4	23.3	19.0	15.8	7.3	-1.5	(30)
(31)		5%	WB	-7.0	-3.9	1.3	6.0	9.2	13.1	15.4	14.8	10.6	5.7	-0.4	-6.3	(31)
(32)			MCDWB	-3.9	1.0	9.1	16.5	22.0	23.2	24.2	23.0	19.1	14.4	5.6	-3.0	(32)
(33)		10%	WB	-8.6	-5.2	0.1	5.0	8.4	12.4	14.6	13.9	9.7	4.5	-1.6	-7.5	(33)
(34)			MCDWB	-5.8	-0.9	7.2	15.1	20.3	23.2	23.8	23.0	19.0	13.0	3.5	-4.6	(34)
(35)	Mean Daily Temperature Range	MDBR		9.0	9.5	10.3	10.0	10.2	9.7	9.3	9.1	9.3	9.5	8.7	8.3	(35)
(36)		5% DB	MCDBR	10.3	10.9	11.5	11.2	11.3	10.8	10.6	10.3	10.3	10.5	10.3	9.4	(36)
(37)			MCWBR	7.4	7.3	6.6	6.0	5.4	4.3	3.7	3.8	4.2	5.6	6.1	6.8	(37)
(38)		5% WB	MCDBR	9.9	10.6	11.0	10.6	10.4	9.5	9.3	8.9	9.3	10.1	9.8	9.3	(38)
(39)			MCWBR	7.3	7.2	6.6	5.9	5.3	3.9	3.8	4.0	4.3	5.5	5.9	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.265	0.303	0.345	0.387	0.402	0.418	0.409	0.390	0.356	0.317	0.283	0.262	(40)
(41)		taud		2.350	2.253	2.132	2.067	2.081	2.113	2.192	2.228	2.278	2.323	2.359	2.351	(41)
(42)		Ebn at Noon		878	889	887	875	870	854	858	863	868	862	844	844	(42)
(43)		Edh at Noon		79	105	135	157	159	155	141	132	116	96	78	72	(43)
(44)	All-Sky Solar Radiation	RadAvg		2.21	3.28	4.93	6.27	7.21	7.15	6.76	6.19	5.29	4.00	2.62	1.99	(44)
(45)		RadStd		0.21	0.29	0.30	0.28	0.32	0.44	0.33	0.24	0.20	0.11	0.16	0.17	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+56

Nomenclature: See separate page